2016

COMPUTER SCIENCE AND ENGINEERING

Paper – CSE – 103

(Advanced Database Systems)

Full Marks – 100

The figures in the margin indicate full marks

Candidates are required to give their answers in their own words as far as practicable

Answer ***any five*** questions

1. Justify on the following comments (***any four***) : 5×4

(a) Data in multiple sites is updated in a pipelined fashion for Hadoop File System (HDFS).

(b) Spatial databases are a variation of distributed database.

(c) Dicing does not necessarily reduce dimensions of a data cube.

(d) A fixed schema may not be very efficient for a semi-structured database application.

(e) Allocation schema is independent of the fragmentation schema.

(f) Normalization may increase time required to process a query for RDBMS.

2. (a) What is a predicate? Give an example.

(b) Define min-term predicate.

(c) Briefly describe the methodology for identification of primary horizontal fragments for the design of a distributed database system.

(d) What is derived fragmentation? Give a suitable example.

(e) In the context of distributed database design, why is it important to identify which part of application data are read-only items? 3+2+7+4+4

3. (a) Draw and describe the different layers for a distributed database architecture.

(b) Explain why the top 3 layers are called site independent layers.

(c) Give an example to show role of semi-join for query optimization in distributed database. 10+5+5

4. (a) What is distribution transparency? How is it achieved for distributed databases in multiple levels?

(b) State two main disadvantages of distributed databases.

(c) Explain the role of local mapping schema towards integration of a heterogeneous multi-site database environment.

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(d) Explain with an example, how a single updation operation on a single data item could result in multiple updation operations on several sites when the database is physically distributed. 5+4+6+5

5. (a) Define the following terms in the context of HDFS :

(i) Name Node (ii) Replication Factor

(b) Why granularity of data is an important aspect towards efficient designing of a data warehouse?

(c) What is data warehouse?

(d) What is Snow-Flake schema? How is it different from a Star Schema for data warehouse?

(e) A cube model has data in four dimensions for a particular subject orientation. The cardinalities for the four dimensions are 3, 3, 5 and 4 respectively.

(i) How many Fact and Dimension Tables are required in a equivalent Star schema?

(ii) State and explain the minimum number of tuples that would occur in each of Dimension and Fact Tables in your design. 4+3+1+4+8

6. (a) What is virtual data warehouse? Why is it gaining importance?

(b) Discuss the 2-phase locking protocol for transaction management in a distributed database environment.

(c) What would be the impact of different granularity of lock sizes for concurrency control in a distributed database environment?

(d) "Normalization is not practiced for designing distributed database systems" — Comment on the correctness of the statement and justify your opinion. 3+8+4+5

7. (a) Discuss the 2-phase commit protocol for transaction management for distributed database.

(b) Define distributed transaction manager (DTM) and local transaction manager (LTM).

(c) Explain if the 2-phase commit protocol is consistent with the ACID properties for transaction management.

(d) What do you mean by Isolation property for databases? 8+4+6+2

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